

OPEN DATA DETECTOR

Technical Design Report

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SELECTED BALANCED TRACKER BASELINE · 2026-10-29

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Source: the detector technical-report repository. The tracker review sections use bundled, hashed parameter snapshots; the text explains the design choices and cites public literature.

Emblem: the Open Data Detector logo, wordmark removed.

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CHAPTER 1

Introduction

This report documents the selected nODD silicon tracker as a baseline for technical review. It develops the pixel, short-strip and long-strip assemblies from their sensors and local structures through barrel/endcap placement and common service interfaces. The tracker chapter preserves both successful model checks and the engineering and coverage cases that remain unresolved.

The report is a living design specification. Public detector experience informs its technologies and construction concepts; nODD dimensions and modeling choices are explained separately. The present review edition does not claim complete detector integration, construction qualification or an accepted performance release. Other detector chapters will be developed against their own reviewed subsystem specifications.

CHAPTER 2

Tracker

2.1 Full-tracker baseline

The tracker combines three silicon technologies in a common cylindrical layout: pixels close to the beam pipe, short-strip strixels at intermediate radii, and long-strip stereo pairs at the outer radii. This edition fixes the selected layouts, local supports and service reservations as a baseline for technical review. The dimensions below are nODD design choices. They describe the selected models, including their engineering approximations; they do not constitute construction approval or a demonstrated tracking-performance specification.

The pixel system supplies two-coordinate measurements for vertexing and forward tracking. The short-strip sensor also measures two coordinates, using a fine transverse pitch and a coarser second coordinate. The outer system uses two separate one-dimensional strip sensors at a small relative stereo angle. Its larger lever arm is intended to constrain the track curvature while controlling channel count. The long-strip pair needs a trajectory model to combine its separated measurements; two sensor faces are not interchangeable with a single two-coordinate plane.

2.1.1 Layout and coordinate conventions

The beam direction defines z, and $r = \sqrt{x^2 + y^2}$ is the transverse radius. The nominal interaction point is at the origin. The selected architecture has ten barrel assemblies: four pixel, four short-strip and two long-strip layers. Each signed end contains ten pixel discs, six short-strip discs and six long-strip discs, giving 22 disc assemblies per end. A long-strip assembly contains both faces of every stereo pair. Figure 2.1 and Table 2.1 distinguish assembly counts from sensitive-element counts.

Pixel barrel datums are at $r = 31.5, 57, 96, 182$ mm, with nominal active extent $|z| \leq 550$ mm. Short-strip stave anchors lie at $r = 260, 340, 480, 660$ mm with nominal active extent $|z| \leq 1\,200$ mm. Long-strip anchors lie at $r = 840, 1060$ mm, with an inherited nominal active half-length of 1 287.33 mm. These radii are datums, not average sensor radii: tilt, module elevation, passive stacks and stereo separation give each sensitive element its own position and normal.

Disc support datums extend to 3 070 mm for pixels, 3 080 mm for short strips and 3 120 mm for long strips. The first short-strip disc is at 1 335 mm and the first long-strip disc at 1 470 mm; both were moved outward to clear their barrel collectors. The associated old-reference coverage losses are retained in the subsystem sections. The geometry is

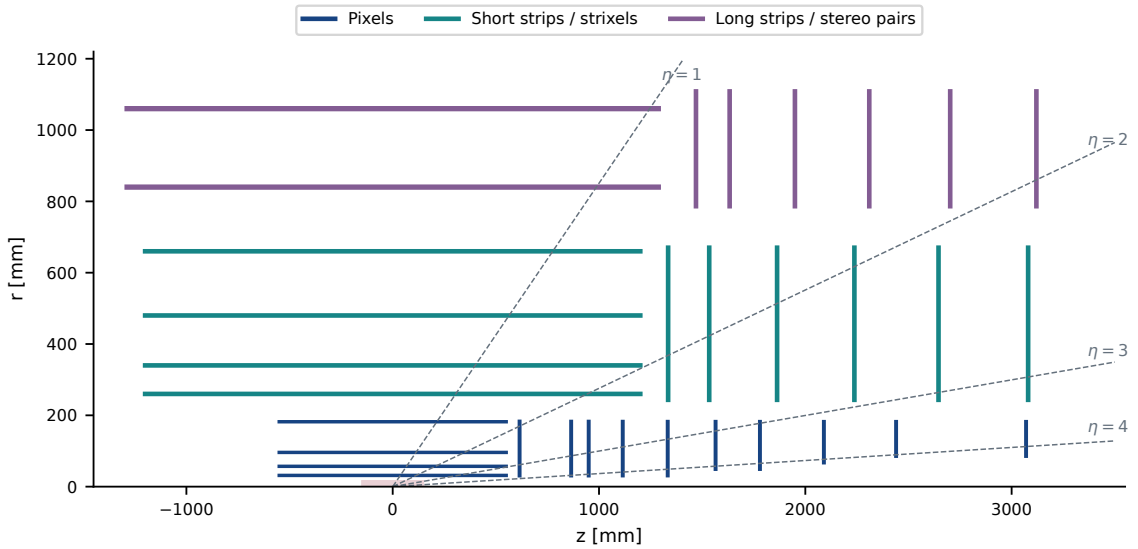


Figure 2.1. Selected tracker layer and disc datums in the r–z plane. The complete barrel and positive discs are shown; the negative end has corresponding signed placements. The lines mark nominal layers and active annuli, rather than all staggered sensor planes or passive volumes. Dashed rays are straight trajectories from the nominal interaction point, not coverage predictions in the field. The red origin interval illustrates the luminous-region hypothesis.

Table 2.1. Selected tracker inventory. Endcap totals include both ends. Pixel chips are active readout patches; strixels are complete two-coordinate sensors; long-strip faces are separate one-dimensional sensors. The maintained full tracker contains 46 250 sensitive elements.

Assembly	Layers/discs	Modules/pairs	Sensitive inventory
Pixel barrel	4	3 050	6 794 chip patches
Pixel endcaps	2×10	2 616	5 616 chip patches
Short-strip barrel	4	8 064	8 064 strixels
Short-strip endcaps	2×6	4 320	4 320 strixels
Long-strip barrel	2	6 984 pairs	13 968 strip faces
Long-strip endcaps	2×6	3 744 pairs	7 488 strip faces

therefore a coherent review layout, with integration and transition-performance questions still open.

The short-strip barrel uses a common +15° azimuthal tilt and the long-strip barrel a common +12° tilt. The complete cold structure rotates with its sensors, tubes and local services. This removes the alternating-radius stave lanes of the earlier short-strip layout and makes the local geometry covariant under a one-stave rotation. Different layer populations and sector services still produce an azimuth-dependent material distribution. Neither this symmetry nor geometric overlap establishes uniform resolution. In the endcaps, module axes follow the local polar frame rather than global x–y directions. Proper rotations define the negative-end strip frames and their identifiers.

2.1.2 Coverage hypotheses and measurement model

The architecture targets charged-particle tracking to $|\eta| < 4$. The pixel active apertures retain the selected ring populations, whose original design screen considered first outward

trajectories with $p_T \geq 1\,\mathrm{GeV}$, a luminous interval $z_0 \in [-150, 150]\,\mathrm{mm}$ and the stated uniform axial-field hypotheses up to 4 T. This interval is a design-screen envelope, not a measured beam distribution. It permits progressively larger pixel active apertures at the outer stations while retaining their common mechanical plate. It does not imply that every strip subsystem independently covers $|\eta| < 4$.

The retained strip screens sample straight tracks and vacuum helices at selected momenta, fields, vertices and azimuths. They use finite sensor planes and retain the original ideal-cylinder or annulus denominators. They are neither proofs for every intermediate momentum nor simulations of scattering, energy loss, dead channels or track reconstruction. Nonzero misses remain at barrel ends and shifted first discs. A finite Fatras truth-tracking comparison uses origin muons at $p_T = 1, 10, 100\,\mathrm{GeV}$, $|\eta| \leq 4$ and a constant 3 T axial field, with geometric digitization as the primary response. About 98.2 % of generated tracks have at least ten physical measurements. This integrated fraction retains forward gaps; it is not a minimum-hit guarantee throughout the acceptance. The selected geometry also has a low-momentum resolution tradeoff. Detailed residuals, pulls and material/response assumptions belong to the accompanying validation evidence.

Pixel readout uses square 50 µm cells. The short-strip fixture uses 75 µm by 0.5 mm cells, with no added stereo pair. Long-strip sensors use 80 µm one-dimensional pitch and physical axes at ±20 mrad about the nominal module direction. These are segmentation and geometry choices. Charge sharing, Lorentz drift, alignment covariance and calibrated spatial resolution remain separate response-model requirements.

2.1.3 Support hierarchy and shared services

The load path is sensor and electronics, local thermal contact, structural sandwich, bearing or petal contacts, then subsystem frames and the global tracker support. In the long-strip system the cooled sandwich is itself the stave or petal; a second hidden carrier is not added between the stereo sensors. The pixel endcaps share a closed carrier with longitudinal rails. Strip barrels use bearing rings, and strip discs use removable petals located by constrained and sliding contacts. These local constructions define load transfer and thermal motion concepts. A qualified global cage, installation tooling and connections to the surrounding detector are still required.

All three systems route local LV, HV and signal services to accessible end boards or rim interfaces, and then to longitudinal trunks. Embedded titanium tubes carry the provisional evaporative CO_2 circuits. Supply, return, cable strain relief and sensor mounting need independent constraints so service loads do not act through silicon bonds. Local loops and common manifolds must also permit pressure testing and subsystem assembly before installation.

The inner pixel trunk reserves $r = 190$–$234\,\mathrm{mm}$ before the rear handoff. The short-strip trunk reserves $r = 710$–$783\,\mathrm{mm}$ from $|z| = 1310$ to $3\,500\,\mathrm{mm}$; the long-strip trunk reserves $r = 1144$–$1169\,\mathrm{mm}$ from $|z| = 1435$ to $3\,500\,\mathrm{mm}$. These fixed envelopes are not expanded when a capacity screen fails. Their separated radial bands alone do not prove installation clearance at collection bays, frames or the rear transitions.

Detailed models contain explicit local sensors, foams, skins, pickups, pipes and selected mounting components. Remote services additionally use effective cells that conserve constituent volumes and residual void. Gross cable area remains a separate packing budget. A combined model must replace the barrel-only strip service cells with the corresponding

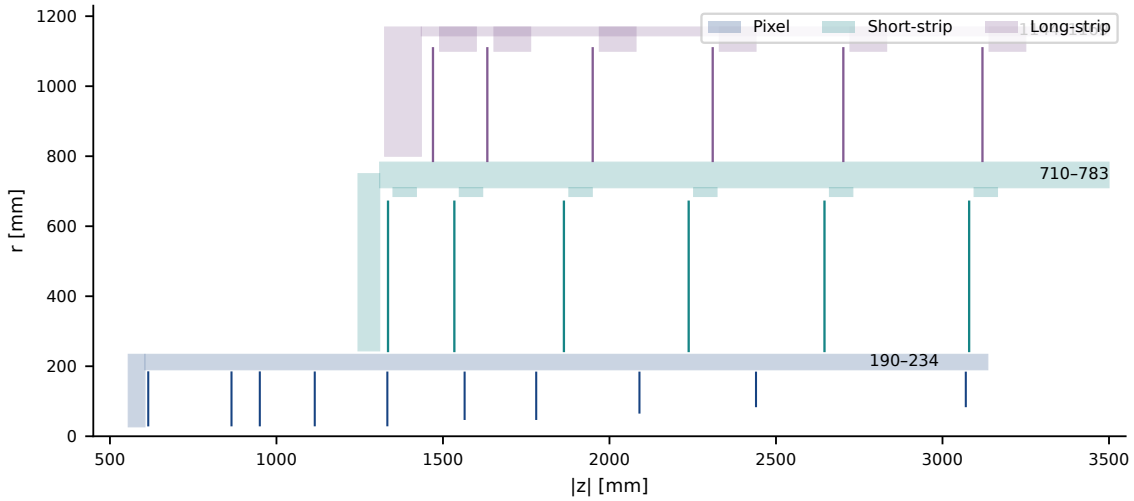


Figure 2.2. Positive-side service reservations at the selected fixed radii. Filled regions show barrel collection bays, disc fans and longitudinal trunk envelopes; they are not filled solids or a cable-installation model. The short-strip envelope is r = 710–783 mm and the long-strip envelope r = 1144–1169 mm. Pixel services occupy their separate inner corridor. Individual fittings, bend sweeps and combined subsystem clearance require review.

cumulative barrel-plus-endcap inventory; overlaying independent compacts would allocate the same corridor and material twice. For the same reason, their standalone model masses must not be summed as a tracker mass or material budget. The maintained integrated exporter reconciles the strip service inventories. Global pixel service capacity and fresh integrated directional material maps remain separate review deliverables.

2.2 Pixel detector

The pixel detector comprises four barrel layers and ten discs at each end, with 5 666 modules and 12 410 active chip patches. The inner two barrels use single-chip modules and the outer two use quads. The endcaps combine four inner single-module rings and two outer quad rings, with a larger active aperture at the forward stations. Carbon sandwich staves and discs carry the modules, cooling and local services. Figure 2.3 shows the selected layout and its connection to the outer service route.

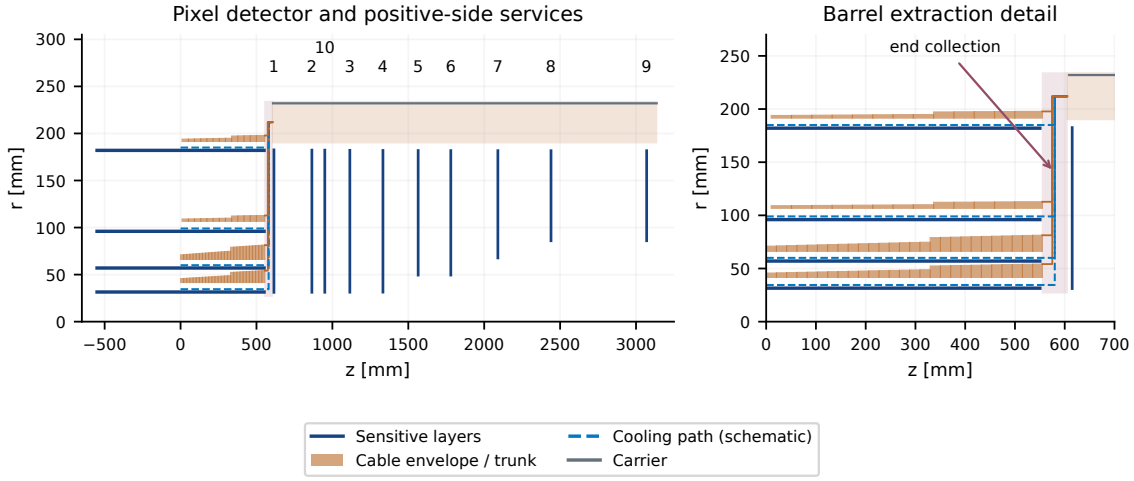


Figure 2.3. Pixel barrel and positive endcap, including local cooling, cable routes and the barrel collection bay. The opposite end is reflected in z.

2.2.1 Modules and readout

Each hybrid module uses a planar n-in-p sensor, ITkPix v2 ASICs and a backside flex. The 400 by 384 matrix has square 50 µm pixels, giving an active area of 20 mm by 19.2 mm and 153 600 channels per chip. The approximate 20 mm by 21 mm die reserves the remaining space for chip periphery.[1] A quad shares one sensor and flex between four active patches, with 614 400 channels. Guards and internal seams are passive silicon. The substrate, active patches and occupied module body are represented separately; their dimensions are given in Table 2.2.

In n-in-p sensors the implants and guard structures face the readout chips; the backside flex carries bias, power and signals. Edge insulation separates the biased sensor from nearby chip surfaces.[2] The flex uses a 25 µm polyimide core and two 12.5 µm coverlays, represented together as 50 µm of polyimide. Two 18 µm copper foils at 50 % coverage and 25 µm epoxy complete its effective material inventory. Sensor and ASIC silicon are each 150 µm thick, separated by a 20 µm bump standoff. Chip peripheries connect to exposed flex pads through the selected 25 µm aluminium bond-wire concept and a module-side termination.

The stack runs from flex copper, polyimide, inner copper and epoxy through sensor, bump standoff, ASIC and graphite contact toward the support. The 0.255 mm graphite shims connect the ASIC backs at $w = 0.245$ mm to the body-back plane at $w = 0.5$ mm. Insulation separates the module's serial-power potential from conductive supports. Figure 2.4 combines the module footprints and stack; the occupied body is 1 mm deep. Endcap

single bodies are 20.4 mm tangentially by 21.4 mm radially; quad bodies are 43.2 mm by 44.2 mm, with ASIC peripheries at the outer radial edges.

Table 2.2. Physical sensor and active-map dimensions in local (u, v), in mm. Active bounding spans include the quad's internal seams; active area is the sum of the separate patches.

	Single barrel	Quad barrel	Single endcap	Quad endcap
Active bounding span	19.2×20.0	38.6×40.2	20.0×19.2	40.2×38.6
Sensor substrate	20.2×20.4	39.6×40.6	20.2×19.4	40.4×38.8
Guard per outer edge	0.5, 0.2	0.5, 0.2	0.1, 0.1	0.1, 0.1
Active area [mm^2]	384	1536	384	1536

2.2.2 Barrel layout

The four layer datums are at 31.5 mm, 57 mm, 96 mm and 182 mm. Single-chip staves tile the smaller circumferences; quads reduce repeated module edges at larger radii. Adjacent staves alternate radial levels, with the offsets and populations in Table 2.3. Their modules have no azimuthal tilt. The 82 staves carry 3 050 modules and 6 794 chip patches.

Local u is tangential, v follows z, and w points radially outward. The 20 mm active dimension follows z, placing the approximately 1.8 mm chip periphery at the transverse edge. The longitudinal guard and body gap are each 0.20 mm. Single/quad body lengths are 20.4 mm and 40.6 mm, with pitches 20.6 mm and 40.8 mm; neighbouring active edges are separated by 0.60 mm, and quad chip seams are 0.20 mm. The selected slim-edge geometry follows the planar-sensor development motivation.[2]

Rows have constant pitch within the 550 mm half-length allocation, with one chip centre at $z = 0$ and a common longitudinal phase between staves. The single-module body is displaced 0.9 mm tangentially from its active centre. Figure 2.5 shows the layer arrangement and packing.

Table 2.3. Selected barrel population. Radial offsets are rounded display values; the printed precision does not specify manufacturing tolerances. Rows means modules per stave. All dimensions are in mm.

Layer	Datum r	Offset ±	Staves	Rows	Modules
B1	31.5	2.492	12	53	636
B2	57	1.692	22	53	1166
B3	96	2.778	18	26	468
B4	182	1.877	30	26	780

2.2.3 Endcap layout

The six-ring pattern contains 102 singles and 50 quads, covering eight radial chip rows. Smaller modules follow the inner curvature; quads share support and service interfaces at larger radii. The local axes are tangential u and radial v around every disc. Module centres account for their individual mounting face and sensor depth. Table 2.4 gives the nominal ring parameters.

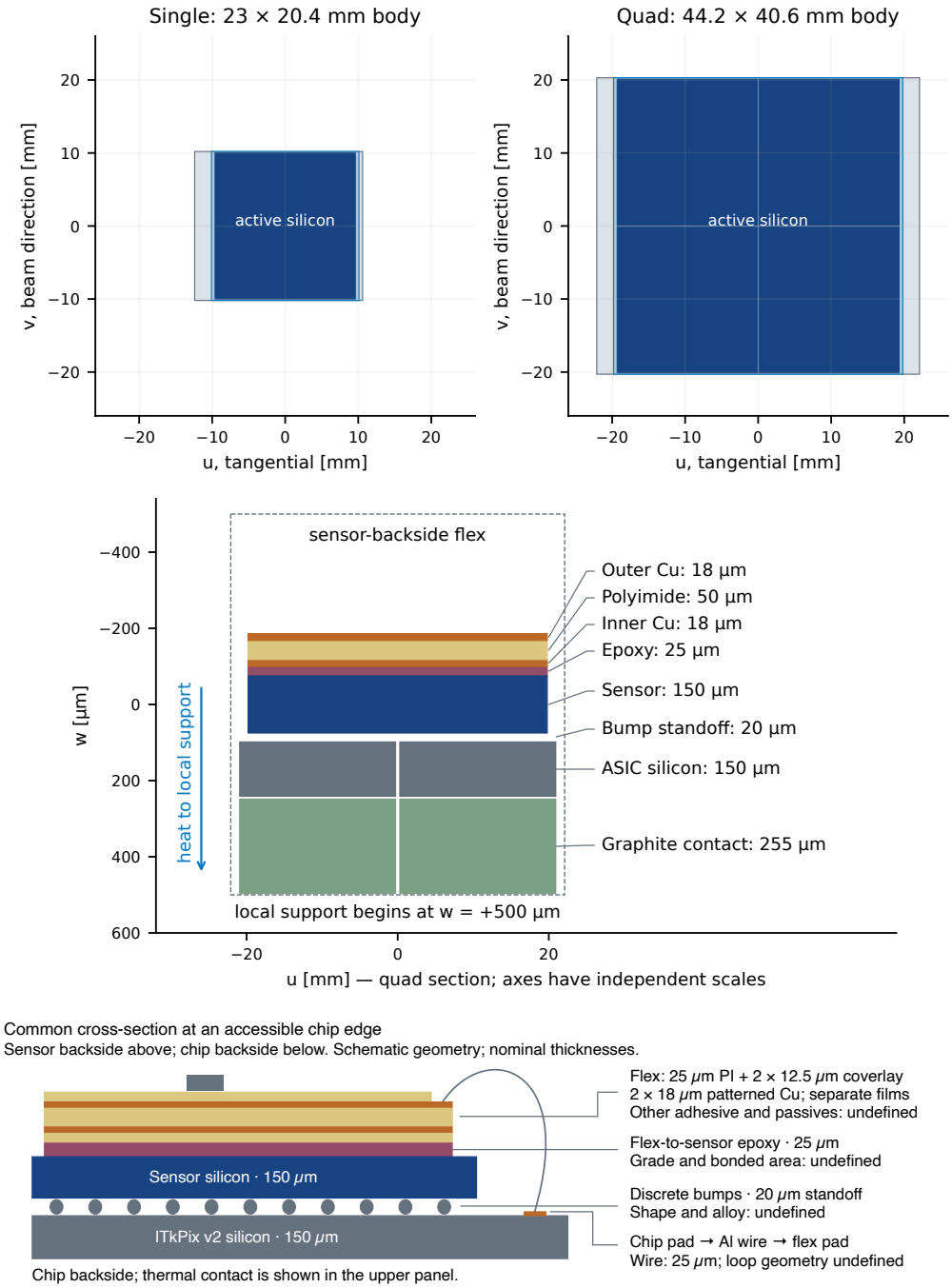


Figure 2.4. Module footprints (top), the quad material stack through one chip row (middle), and schematic bond access (bottom). Active patches, passive silicon and occupied envelopes are distinct. Stack thicknesses use an independent scale; the electrical detail illustrates discrete connections.

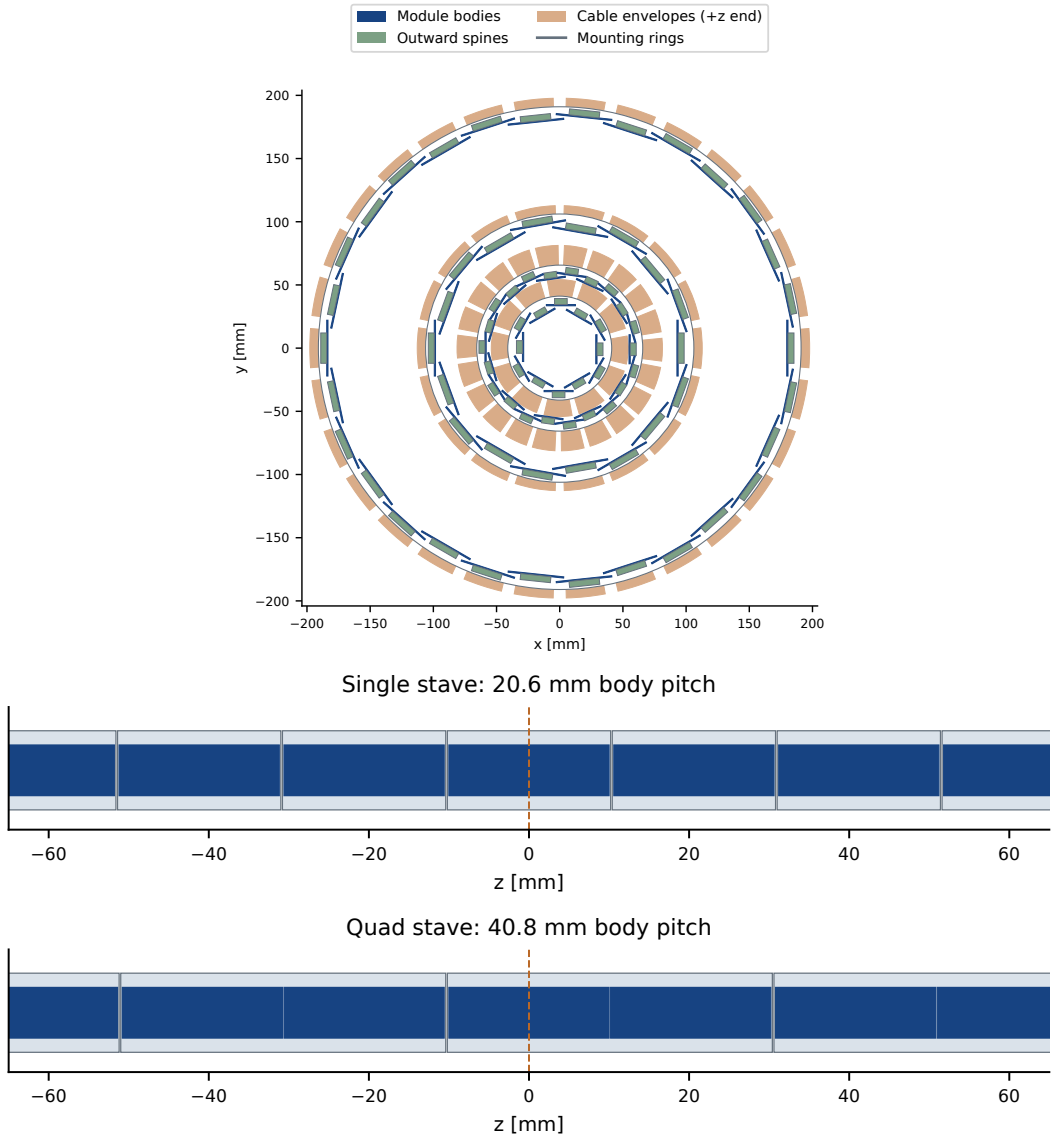


Figure 2.5. Transverse barrel assembly projection (top) and central single/quad row packing (bottom). Module bodies overlap tangentially on staggered radial levels; cable envelopes pass outside the bearing rings. The lower panel resolves active intervals, guards and gaps about $z = 0$.

Table 2.4. Six-ring source pattern. Radii are the uncorrected radial placement parameters; actual module centres retain the face/height-dependent transverse compensation. Each ring occupies both mounting faces and two height levels per face. Quad rings contain two radial chip rows.

Ring	Family	Nominal r [mm]	Modules	Chips
1	Single	41.150	18	18
2	Single	59.283	22	22
3	Single	77.500	28	28
4	Single	95.779	34	34
5	Quad	124.000	22	88
6	Quad	161.000	28	112

The station inventory in Table 2.5 specifies the retained inner rings and both outer quad rings at each datum. The larger forward active apertures are implemented by retaining the selected ring populations at each station. Local module positions, face assignments and sensor depths are preserved within each rigid disc assembly. Disc identifiers remain stable: disc 10 lies at $|z| = 950$ mm between discs 2 and 3 and repeats the complete local geometry of disc 2. The table is ordered by distance from the interaction point. These apertures and datums do not imply complete acceptance for every trajectory. Both endcaps contain 2 616 modules and 5 616 chip patches. The negative end uses corresponding signed datums and support-facing module frames.

Table 2.5. Selected disc support datums and retained inventory on one side. Ring numbers refer to the six-ring source pattern; quad rings 5 and 6 remain at every station. The negative side uses the opposite datum sign and the same populations. Datums identify rigid assemblies, not manufacturing tolerances.

Disc	$\|z\|$ [mm]	Single rings	Singles	Quads	Chips	Circuits
1	615	1, 2, 3, 4	102	50	302	16
2	865	1, 2, 3, 4	102	50	302	16
10	950	1, 2, 3, 4	102	50	302	16
3	1115	1, 2, 3, 4	102	50	302	16
4	1333	1, 2, 3, 4	102	50	302	16
5	1565	2, 3, 4	84	50	284	14
6	1780	2, 3, 4	84	50	284	14
7	2090	3, 4	62	50	262	12
8	2440	4	34	50	234	10
9	3070	4	34	50	234	10

Modules alternate between the two plate faces and between two heights on each face, keeping overlapping occupied bodies separated in depth. A quad's four patches share one sensor plane. Sensor centres are at local $w = \pm 4.10$ or ± 5.75 mm; ASICs and graphite contacts face the plate. An outgoing track crosses the plate before an outward-facing sensor. The first and last discs are compared in Figure 2.6.

2.2.4 Supports and cooling

Barrel staves use carbon-foam cores, CFRP skins, graphite spreaders and two embedded titanium tubes. The 10 mm single and 24 mm quad spines support wider module-facing plates, bringing heat from the chip area toward the cooled core. The support lies radially outside the sensor. Six bearing rings per layer connect the staves to the barrel cage at $z = \pm 110, \pm 330, \pm 546$ mm, giving 24 rings and 492 feet. The station at $z = -546$ mm sets the axial datum; the remaining contacts guide the stave while allowing longitudinal contraction. Two independent evaporators per stave run in opposite directions between its ends. Carbon sandwiches and titanium cooling tubes have public pixel stave precedents.[3]

Each endcap plate remains a full annulus from $r = 27$ to 188.5 mm, including on discs with fewer active rings. These dimensions and the routing ledger describe the standalone pixel model. In the integrated tracker, passive inner apertures are clipped to 28.8 mm, giving 1 mm clearance to the 27.8 mm outer beam pipe. Non-filler service and cooling inventory is conserved by the integrated material representation. A per-chip pickup and

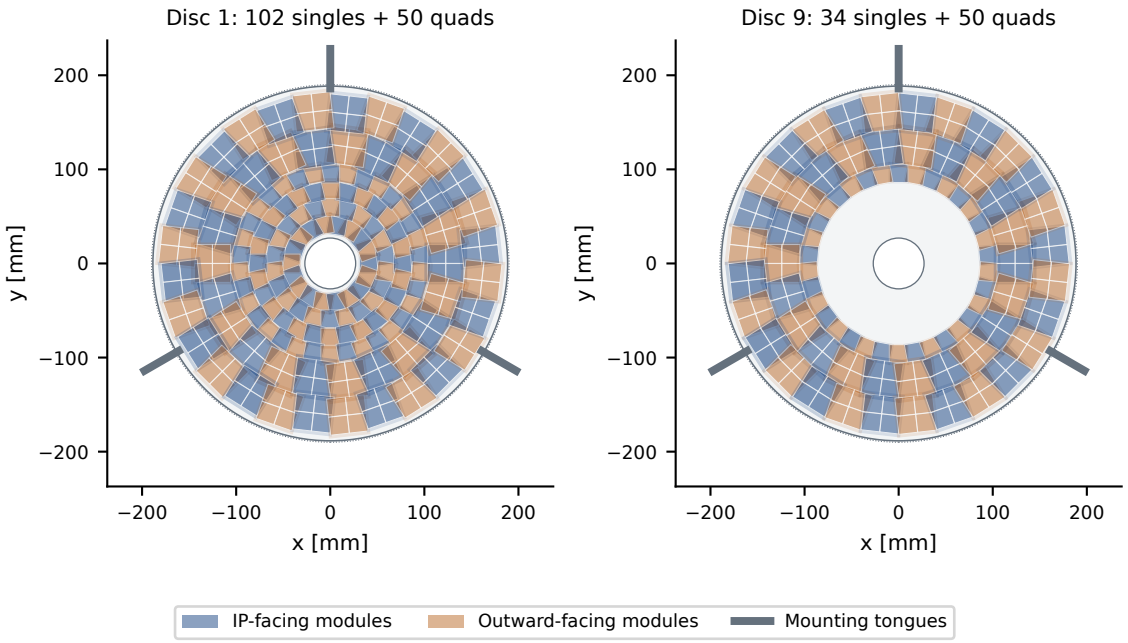


Figure 2.6. First and last positive discs at a common scale. The first has four single rings and two quad rings; the last retains single ring 4 and both quad rings. Colour distinguishes face and height. The views project all sensor depths onto the transverse plane.

short graphite stem connect each ASIC to the sandwich and titanium tube. Graphite orientation defines the thermal-conduction path.[4, 6] Azimuthal and radial tube routes occupy separate planes at $w = \pm 1.5\,\mathrm{mm}$. Each chip row has two independent half-ring circuits; the station totals are in Table 2.5, giving 280 circuits across the two endcaps. Three rim tongues transfer loads through cone/slot/plane contacts to box rails on a common CFRP carrier. Table 2.6 and Figure 2.7 specify the local passive geometry.

The barrel tube centres are tangentially at ±2.2 mm for inner staves and ±6 mm for outer staves. The 164 full-length tubes join 82 feed and 82 return branches at each signed end. Thin titanium circuits with evaporative CO_2 provide the reference cooling topology.[5, 6]

Table 2.6. Support and mounting ledger. Lengths are in mm. Internal adhesive and inserts replace core material within the stated envelope.

Component	Selected dimensions
Stave stack, module outward	Interface 0.100; insulation 0.025; graphite 0.200; CFRP 0.150; foam 4.100; CFRP 0.150; total 4.725. Internal adhesive equivalent 0.080.
Barrel bearing rings/feet	Ring axial width 8, radial depth 1; reference corner clearance 0.5. Rings follow layer translations; current gaps are 0.474, 0.488, 0.440 and 0.500. Foot width 4, axial length 8; height follows stagger.
Passive stave span	Singles: $z = -551.0$ to $+552.8$; quads: $z = \pm 552.8$.
Disc sandwich	Core 6.0; each CFRP skin0.15; total 6.3.
Per-chip pickup	Area 384 mm^2; graphite 0.30; cradle 0.075; insulation/interface/adhesive0.025 each; total 0.45.
Disc stem and inserts	Stem section 6×8; heights 0 or 1.65; quad stems shift 1 toward module centre. Core insert 4.50; skin window 0.15; contact area 48 mm^2; adhesive equivalent 0.08.
Disc tongues	Width 10; $r = 186$–228 at $\phi = 90, 210, 330°$.
Carrier rails/shell/flanges	Station-following coupling slots; three 8×8 box rails, wall 0.4 at $r = 223.7$–231.7; shell $r = 231.7$–232.0, $\lvert z \rvert = 609$–3136; end flanges radial 10, axial 2.

Table 2.7. Reference cooling inputs. Diameters/walls are in mm; flow is per independent tube or circuit. The common heat allowance is 2.688 W per chip; the endcap flow budget uses a factor 1.5 stress load.

Location	Ti diameter	Ti wall	CO_2 [°C]	Flow [g s^{-1}]
Inner barrel	2.0	0.11	−35	1.3
Outer barrel	2.8	0.15	−35	2.5
Endcap, inner chip rows	2.8	0.15	−40	1.0
Endcap, outer quad chip rows	2.8	0.15	−40	1.1

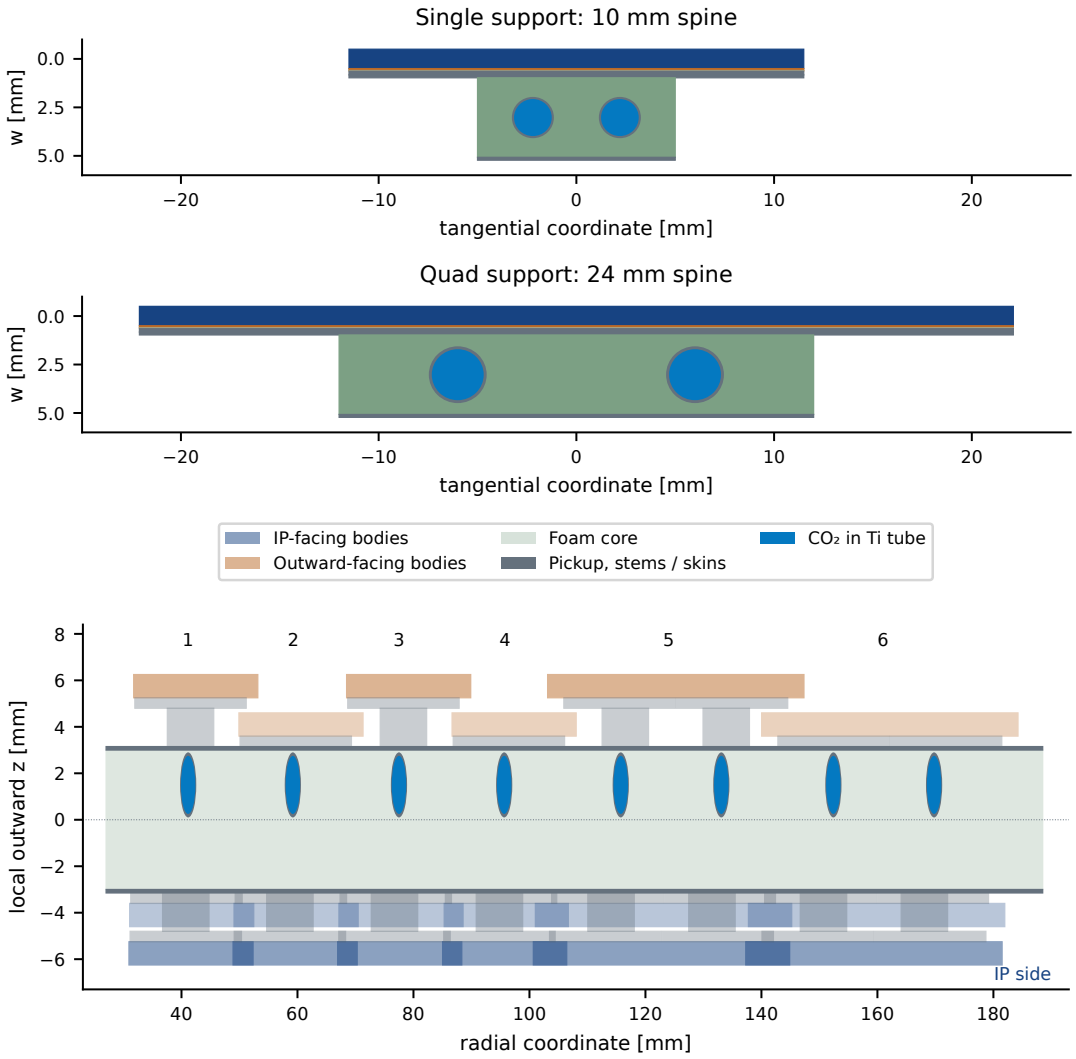


Figure 2.7. Stave sections (top) and a radial/axial disc projection (bottom). Carbon, foam, titanium and coolant form the passive thermal path. The disc panel superposes both mounting faces and heights; short pickups connect the chip backs to the plate. Thin layers are specified in the support ledger.

2.2.5 Services and material representation

Barrel electrical services follow each half-stave toward its nearest end; the central single-chip module joins the positive half. Two serial chains serve each half-stave, with limits of 16 modules and 32 chips, giving 164 chains at either end. Cable envelopes grow with the cumulative module and chain load and pass outside the bearing rings. Cooling and electrical services collect in twelve sectors in the $|z| = 555$–605 mm bay, then join the axial reservation near $r = 212$ mm. Two sectors serve six staves each and ten serve seven each. Figure 2.8 illustrates the local route; Table 2.8 records its envelope rules.

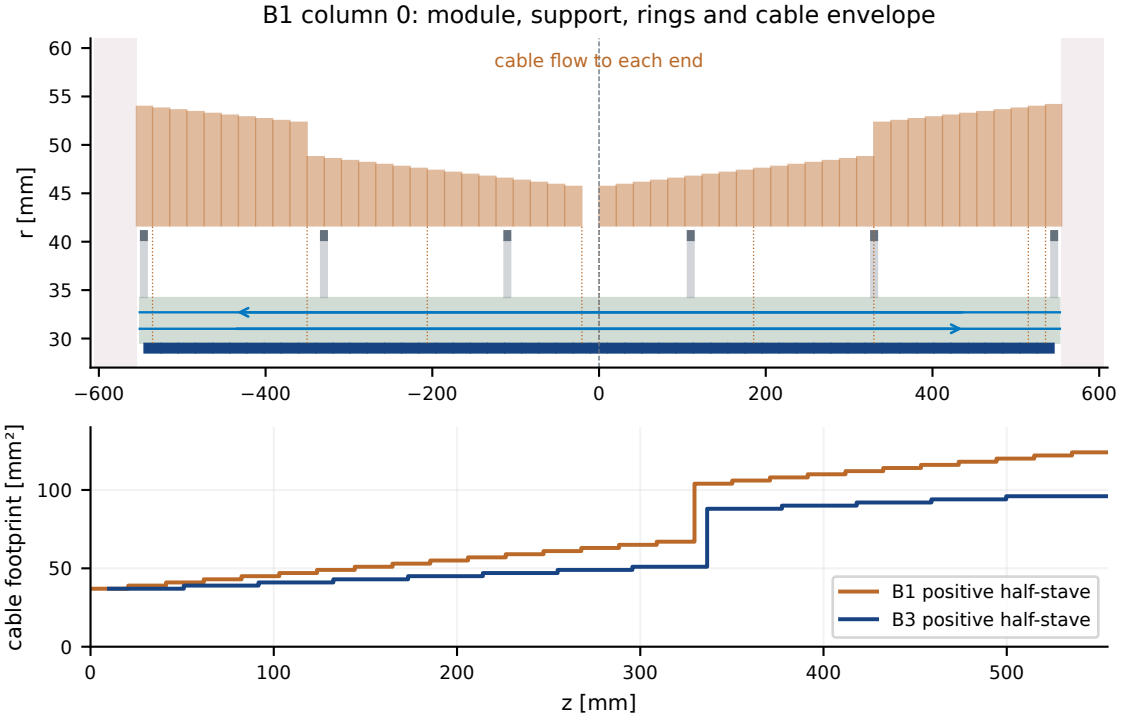


Figure 2.8. Barrel service routes and cumulative cable footprint. The upper panel projects the occupied bodies, support, bearings and conceptual pickups. Counterflow paths are separated radially for display; their physical tubes are separated tangentially in the core.

Endcap tails join ring buses on their own module face. Three radial fans per face gather them at the disc rim. Electrical chains stop at half-ring boundaries and contain at most 32 chips, independently of the cooling circuits: the full disc has 18 chains, reducing to 16, 14 and 12 on the forward stations. A 50 mm-deep collector carries each load into the axial trunk. The last disc uses an inner bypass to the rear turn at $|z| = 3\,150$ mm. Figures 2.9 and 2.10 show the face routes and effective transport cells.

The simulation separates sensitive silicon from passive module bodies, supports and services. Explicit sensor substrates, chip patches, ASICs, graphite contacts, stave spines, tubes, rings and feet preserve material ordering around the measurements. Effective components conserve their constituent inventory as summarized in Table 2.9.

Remote cable footprint uses provisional volume fractions of 10 % copper, 40 % polyimide and 50 % air, with a separate 50 % inter-cable packing allowance. CO_2 inventory uses a reference density of $1.096\,4$ g cm^{-3}, the saturated liquid density at -35 °C.[7] The standalone pixel model-inventory mass is 204.747 kg, including 159.749 kg in downstream transport cells. The maintained DD4hep geometry repeats the detailed disc assemblies, including

Table 2.8. Electrical and remote-service ledger. Lengths are in mm, areas in mm^2. Cable footprint and inter-cable packing are separate budgets.

Component	Allocation and route
Barrel cables	One data and one command link per module, footprint 1 each; ancillary footprint 35 per chain from its first pickup. Sector width 75% of stave pitch, packing 50%; inner radius 0.5 beyond the outer bearing ring.
Half-stave terminal loads	Singles: ancillary 70 plus links 52 at $-z$,54 at $+z$; quads: ancillary 70 plus links 26.
Grouped coolant transport	Feed and return branches reserve diameter4; a grouped leg for N staves has envelope diameter $4\sqrt{N}$.
Endcap face flex	Ring buses 8×0.20; module tails 8×0.12; three radial fans per face.
Disc collector	r = 27–190; local outward z = 11.7–61.7 from its datum.
Axial trunk	Reservation r = 190–234; material cells r = 192–231.7. Three 90° sectors separated by 30° mounting sectors centred at $\phi = 90, 210, 330°$. Barrel load enters at $\lvert z\rvert$ = 605; disc collectors follow their station datums; the rearmost disc is logical disc 9.
Rear interfaces	Last collector ends at $\lvert z\rvert$ = 3131.7; bypass r = 27–188 to rear turn $\lvert z\rvert$ = 3150. Rear cell extends to $\lvert z\rvert$ = 3300, r = 680. Flange necks at $\lvert z\rvert$ = 609–611 and 3134–3136 have outer radius 222.

Table 2.9. Selected representations and the physical quantities they preserve. The effective components remain identifiable material allocations.

Component	Representation and scope
Disc sandwich	Explicit annulus and disjoint tongues; constituent-normalized core for inserts, adhesive and closeouts.
Cooling	Explicit Ti/CO_2 evaporator tori; summed half-ring length preserved. Radial legs and bends retain effective inventory.
Module pickup	Explicit per-chip pickup stacks and short stems; contact windows and cradle openings preserved through effective volumes.
Disc flex and fittings	Local effective annulus beyond module envelopes; flex, clip, coupling and bend allowances remain separate inventory terms.
Collector and trunk	Effective transport cells preserve conductor, insulation, titanium and coolant volumes through service handoffs.
Carrier	Shell, curved box rails and flanges partitioned to avoid counting intersections twice; coupling pockets remain reserved.

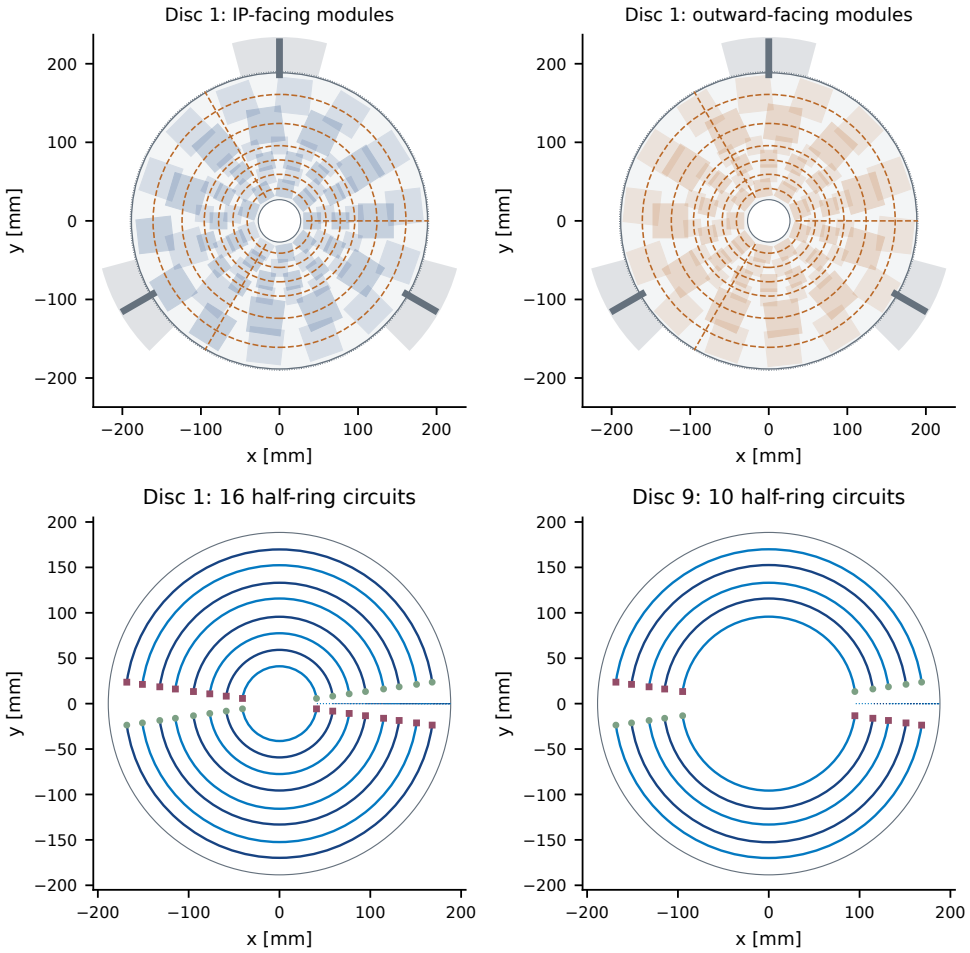


Figure 2.9. Disc face buses and independent half-ring cooling paths. The first/last discs have sixteen/ten cooling circuits. Module outlines preserve their frames; bus lines, fans and inlet/outlet symbols show service topology.

sensors, passive stacks, cooling, mounting tongues and local collectors. Rail coupling slots follow all ten stations; carrier and service envelopes remain fixed. Global trunk, bypass and rear cells retain the reference material inventory. Their cumulative connectivity and capacity for the selected populations, and the thermal, hydraulic and mechanical margins, remain unqualified.

The reconstruction comparison uses a separate effective ACTS material representation: existing material profiles are transported with their assigned surfaces and the added disc has a homogeneous donor-disc material layer behind its sensors. That approximation does not replace the detailed DD4hep assembly. A new Geant4 material map and candidate-specific response validation remain separate requirements.

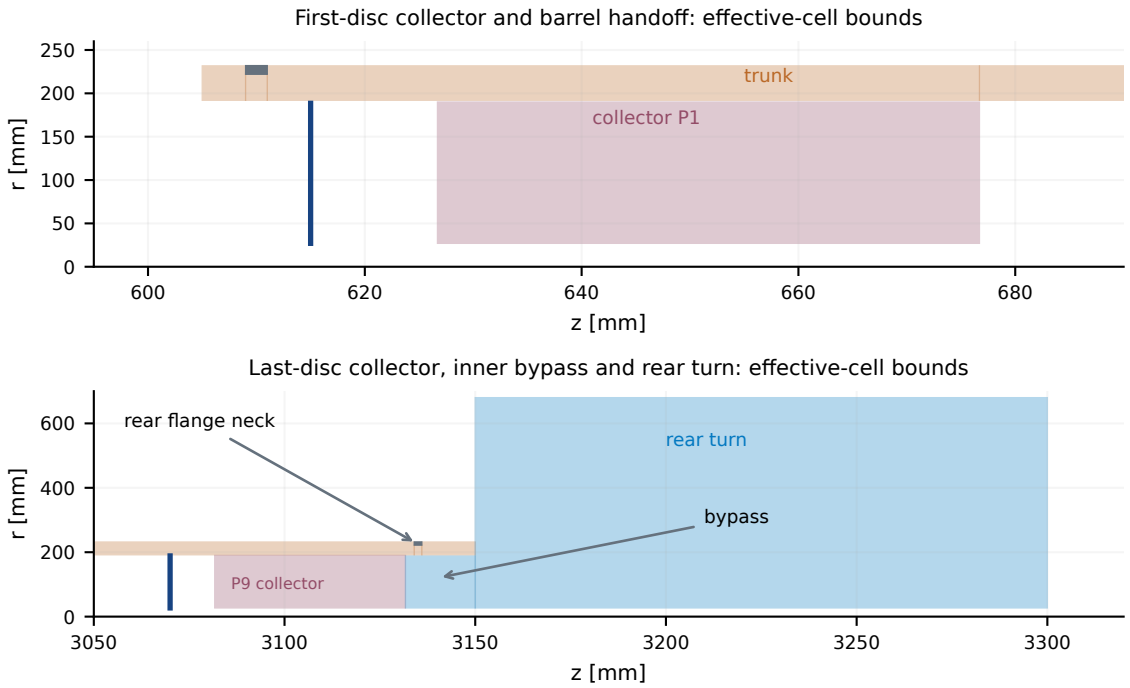


Figure 2.10. Effective collection, trunk, final-disc bypass and rear-turn cells. Carrier-flange necks retain their fixed bounds. Collector cells follow the station datums; the trunk and rear cells retain the reference constituent inventory within their fixed bounds.

Pixel detector references

[1] Stefano Esposito, on behalf of the RD53 Collaboration, *RD53: Lessons Learned — A verification perspective*, 2023-10-23, slide 5: device generations, ATLAS matrix and approximate die outline. Public presentation.

[2] Philipp Weigell, on behalf of the ATLAS Planar Pixel Sensors R&D Project, *Recent Results of the ATLAS Upgrade Planar Pixel Sensors R&D Project*, arXiv:1210.7661v2 (2013). Section 3.1.1, PDF p. 2: n-in-p junction, guard and edge-insulation interfaces; §3.2.1, PDF p. 4: slim-edge sensor developments. Public paper.

[3] ATLAS IBL Collaboration, *Production and Integration of the ATLAS Insertable B-Layer*, JINST 13 (2018) T05008, arXiv:1803.00844v3. Section 4.1, Tables 9–10, PDF pp. 38–39: carbon supports and their constituents; §5.2.1, Fig. 33, PDF pp. 49–50: titanium cooling; §7.3, PDF p. 75: ring/foot mounting interfaces. Published article; public preprint.

[4] Richard J. Lemak, Robert J. Moskaitis, Yoshiro Nogami and Katsuhiro Chikuba, *High Performance Pyrolytic Graphite Composite Heat Spreaders*, IMAPS Advanced Technology Workshop on Thermal Management (2011), slide 6: in-plane and through-plane thermal conductivity. Public presentation.

[5] Sonia Carrà, on behalf of the ATLAS ITk group, *ATLAS ITk Pixel Outer Endcap CO_2 cooling system prototypes*, ATL-UPGRADE-PROC-2024-001, 2024-06-20. Section 1, PDF p. 1: temperature targets; §2–4 and Figs. 1, 3, PDF p. 2: thin titanium circuits, joints and prototype cooling arrangements. Public preprint.

[6] V. Lukashenko, on behalf of the CMS Tracker group, *Results of CMS Phase-2 TEPX system tests*, JINST 21 (2026) C02027. Section 1 and Fig. 1, PDF p. 2: composite/titanium supports; §5, PDF pp. 5–6: module contact force and thermal interface measurements. Published article.

[7] NIST Chemistry WebBook, Standard Reference Database 69, *Carbon dioxide saturation properties*, CAS 124-38-9. Saturation table from −40 to −30 °C in 5 °C steps, −35 °C row, liquid-density column; accessed 2026-09-30. Public tabulated data.

2.3 Short-strip detector

The intermediate tracker uses rectangular two-coordinate sensors, referred to here as strixels. The selected layout combines four tilted barrel layers with six petal discs at each end. It retains a common module contract for barrel and endcap, with the fine coordinate transverse to the track's radial direction and the coarse coordinate longitudinal in the barrel or radial in the endcap.

2.3.1 Sensor, electronics and thermal stack

Each module has a 48 mm by 96 mm active area in a 0.2 mm silicon sensor, with a 0.5 mm guard on every edge. Its 75 µm by 0.5 mm grid contains $640 \times 192 = 122\,880$ nominal cells. Eight 23.4 mm-square, 0.15 mm-thick readout tiles occupy a 2×4 array at 24 mm centre pitch. This is a footprint hypothesis: redistribution, tile gaps, channel compatibility and readout power require a real electronics design. Cell counts do not establish live-channel coverage across the tile boundaries.

The modeled module includes a 25 µm bump envelope, represented by copper and air fractions, followed by 0.10 mm adhesive, 0.20 mm graphite, 25 µm insulation, 0.10 mm adhesive and 0.20 mm CFRP backing. Two 3 mm axial flex ledges enlarge the occupied outline to 49 mm by 103 mm. Their bridges and tails are included in clearance checks. The graphite and local pickups couple the module to the cooled sandwich. Neither the assumed contact resistance nor the carbon conductivity is a qualified cold interface.

2.3.2 Tilted barrel placement

The stave anchor radii are 260, 340, 480, 660 mm, with 44, 56, 80, 108 staves respectively. Each carries 28 modules, giving 288 staves and 8 064 sensors. The modules have symmetric endpoint centres at $z = \pm 1\,152$ mm and 85.333 mm row pitch; their axial active overlap is 10.667 mm. Alternating rows are lifted by 1.5 mm to separate complete occupied modules. The endpoint active edges reach the nominal $|z| = 1\,200$ mm. The tilt angle has been optimized to +15° about global z.

2.3.3 Structural stave, bearings and barrel routes

The 52 mm-wide cold structure extends to $|z| = 1\,210$ mm. It contains a 5 mm conductive carbon-foam core, 0.10 mm core adhesive and 0.15 mm CFRP skins on each face. The skins carry bending load; the core separates them, transfers shear and houses the cooling. Module pickups follow each row's elevation. Both pipes and their coolant volumes are explicit, and their full outer envelopes are subtracted from the core.

Nine CFRP bearing rings per layer are centred at $z = 0, \pm 250, \pm 550, \pm 850, \pm 1150$ mm. Each is 8 mm wide axially and 3 mm radially. A beveled 2 mm-wide web at local $u = 14.5$ mm joins the bare back-skin land to the ring tangent plane, leaving 0.5 mm to the adjacent insulation and signal bus.

Each signed half-stave has one titanium U-loop with two straight legs at $u = \pm 12$ mm, tube outside diameter 2.5 mm, wall 0.14 mm, and centre $w = -4.05$ mm. Straight runs extend from $|z| = 16$ to 1 210 mm; the central return has 12 mm centre radius. Positive and negative circuits are hydraulically separate. The backside LV carrier holds two 12 mm by

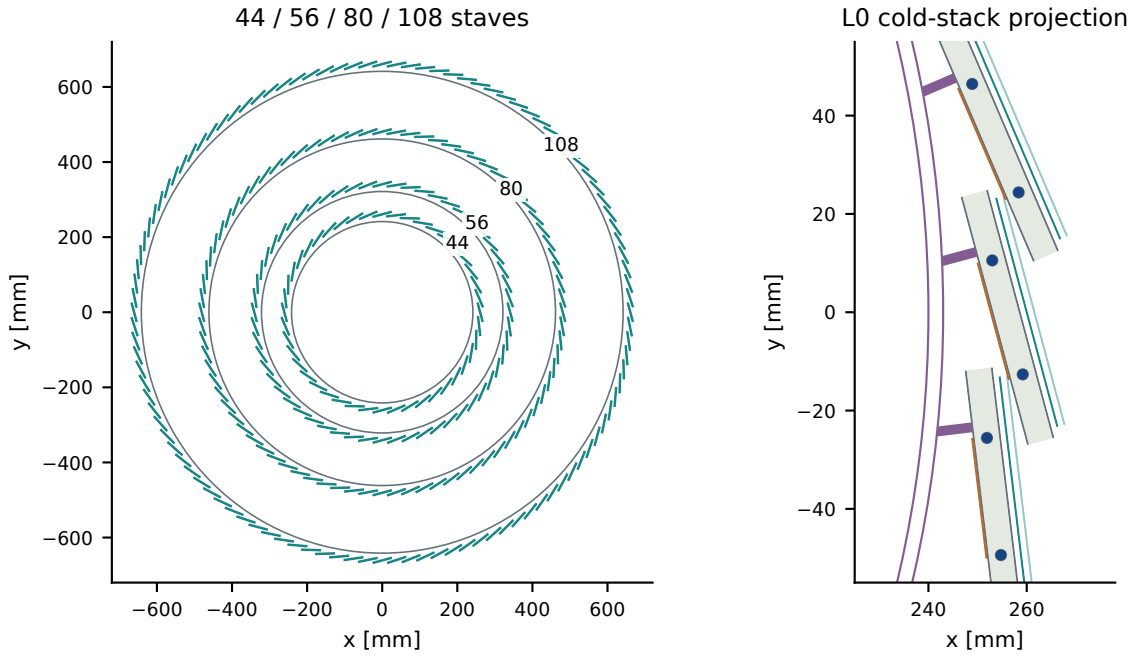


Figure 2.11. Selected short-strip barrel. Left: both row elevations are projected for every stave; labels give layer populations. Right: the innermost cold-stack projection shows equal-radius shingling, embedded tubes and the bearing contact on a bare skin land. It superposes different longitudinal rows, rather than showing one physical z cut. The common azimuthal tilt is 15°.

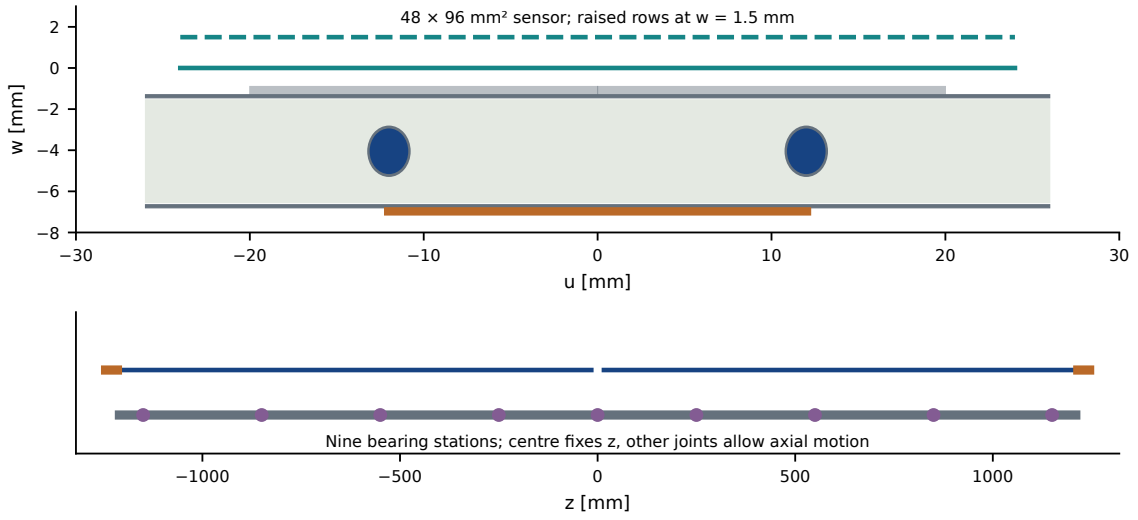


Figure 2.12. Short-strip stave structure and bearing schedule. Top: local layer positions and embedded tube sections; low and raised sensors are superposed. Thin module films and complete pickup details are omitted from this explanatory view. Bottom: the nine selected bearing stations and the two independently served half-staves. End boards lie beyond the active rows.

Table 2.10. Positive support datums for the strip discs. The opposite end uses the corresponding signed placements. Values are rounded for display. The first short- and long-strip datums include the barrel-collector clearance amendments.

Disc	Short strips $\lvert z\rvert$ [mm]	Long strips $\lvert z\rvert$ [mm]
1	1335.00	1470.00
2	1534.17	1633.21
3	1863.16	1949.64
4	2237.84	2310.00
5	2645.64	2702.23
6	3080.00	3120.00

0.20 mm copper rails on 0.15 mm polyimide. HV, control and data use a separate patterned flex. Module tails route around the cold structure to these insulated buses.

End-board allocations are 52 mm by 30 mm by 8 mm, at $z = \pm 1\,230$ mm. Each half-stave's 14 modules require two harnesses under the 12-module limit. Each end therefore presents 576 harnesses and 288 cooling loops, grouped into twelve sector collectors at $r = 244$–750 mm, $|z| = 1245$–1310 mm, before the fixed outer trunk. The gross harness fixture uses 13.4 mm power and 3.6 mm fibre envelopes. These dimensions are capacity comparators, not a manufactured cable specification.

2.3.4 Endcap discs and removable petals

Six discs per end use the datums in Table 2.10. The nominal active annulus spans $r = 243.5$–670.06 mm. Twelve 30° petals divide each disc into removable cold assemblies. Five radial rings carry 48, 60, 72, 84, 96 modules, or 30 per petal and 360 per disc; the two endcaps contain 4 320 sensors. Fine U is tangential, coarse V points inward radially, and N points outward from the interaction region. Alternating ring and azimuth elevations give four offsets 0, 1.5, 3, 4.5 mm. The rectangular bodies overlap in projection while their full passive stacks remain separated in depth.

The petal spans $r = 236$–685 mm with a 0.6 mm inner seam gap. Its 5 mm foam, 0.10 mm glue and 0.15 mm skins form the cooled structural plate. Two 6 mm by 60 mm graphite pickups per module follow the elevations; their small thermal area is included in the adverse screen. Inner and outer CFRP frames occupy $r = 236$–250 and 674–700 mm.

One serpentine loop per petal runs on ten circular tracks at the five module radii plus or minus 20 mm, with radial U-turns between tracks. The feed and return approach the rim at 29 and 28°, through 10 mm entry fillets. The tube dimensions match the barrel. LV and HV/data buses remain separate on the back; front fanout cells preserve flex constituents and approximate length above the modules, without representing every connector, via or vertical transition. The 52 mm by 20 mm by 8 mm outer board sits at $r = 695$ mm. Each petal supplies three harnesses and one loop to its $r = 685$–710 mm outer fan, then to the shared trunk. The endcaps add 216 harnesses and 72 loops per end, giving combined barrel/endcap totals of 792 harnesses and 360 loops per end.

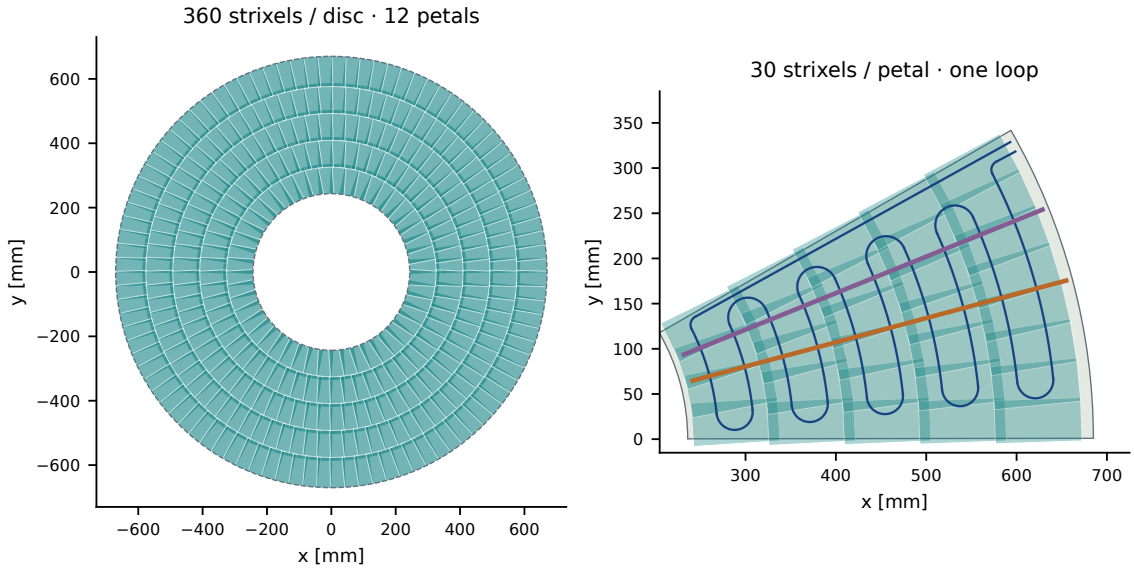


Figure 2.13. Short-strip endcap, with all four elevations projected. Left: the first positive disc's five rectangular module rings. Right: one petal and its embedded serpentine cooling centreline. The two backside bus routes are shown schematically at their selected azimuths. The outer board, fanout transitions and individual connectors require detailed routing review.

2.4 Long-strip detector

The outer tracker uses complete stereo pairs on a continuous cooled structural sandwich. Two tilted barrel layers and six petal discs at each end share a one-dimensional sensor contract. The sandwich between the sensors provides their local mount, thermal path and stave or petal structure; no additional unmodeled rail is credited with carrying the load.

2.4.1 Stereo sensors and module construction

Each face has a 96 mm-square active area in 0.3 mm silicon, with 0.5 mm guard and 1200 strips at 80 µm pitch. The physical in-plane axes rotate by ±20 mrad, giving a 40 mrad relative stereo angle. Each face returns one measured coordinate. Combining a pair must transport both measurements to a common reference plane using the track slope; the finite face separation must not be hidden in a fictitious two-coordinate segmentation.

The barrel core is 5 mm carbon foam with 0.10 mm adhesive and 0.20 mm CFRP skins on both sides. Each sensor couples through 0.20 mm module glue, a full-area 0.10 mm graphite spreader and a 42 mm-square pickup. The pickup is 0.10 mm high on a low row and 1.60 mm on a raised row. Complete pairs are lifted outward on both faces, giving midplane separations of 6.7 mm or 9.7 mm. This replaces the thinner analytical gap with the actual occupied sandwich. Electrical hybrids and insulated buses occupy the opposite edge from the mounting web.

2.4.2 Barrel placement and bearing connection

Stave anchors at $r = 840$ and 1 060 mm carry 86 and 108 staves, respectively. Each stave has 36 complete pairs, giving 6 984 pairs and 13 968 independent sensor faces. The common azimuthal tilt is +12°. The selected 32 mm transverse reserve and 70.765 mm row pitch retain projected overlap for finite stereo separation and curved incidence. They increase silicon and channel inventory substantially relative to the original sparse layout; this is an explicit coverage/material tradeoff. Full finite-plane screens, rather than body projections alone, determine whether both faces of the same pair are hit.

The stave is 124 mm wide and extends to $|z| = 1\,320$ mm. The free shingle edge at local $u = +55$ mm carries its mounting lands; local buses and hybrids lie at $u = -55$ mm. Two potted epoxy/titanium fasteners at every station connect through a keyed CFRP web and tangent shoe to an inner CFRP bearing ring. Rings occupy $r - 45$ to $r - 42$ mm and are 6 mm wide axially. The centre station locates z; the remaining stations propose longitudinal sliding with ±0.3 mm travel. This is an assembly provision for thermal motion and placement allowance, not a detailed machined slot or a qualified clamp/preload specification.

2.4.3 Anchors and sag limits

The structural screen includes the attached core, skins, sensors, thermal parts, electronics, tubes, buses and local mounting load. Grounded bearing rings are excluded from the stave payload; the worst stave carries about 1.8 kg. A released-span bending-and-core-shear calculation uses skin modulus 70 GPa, core shear modulus 5 MPa and twice gravity. These are conservative screen hypotheses within the tested range, not measured material lower bounds.

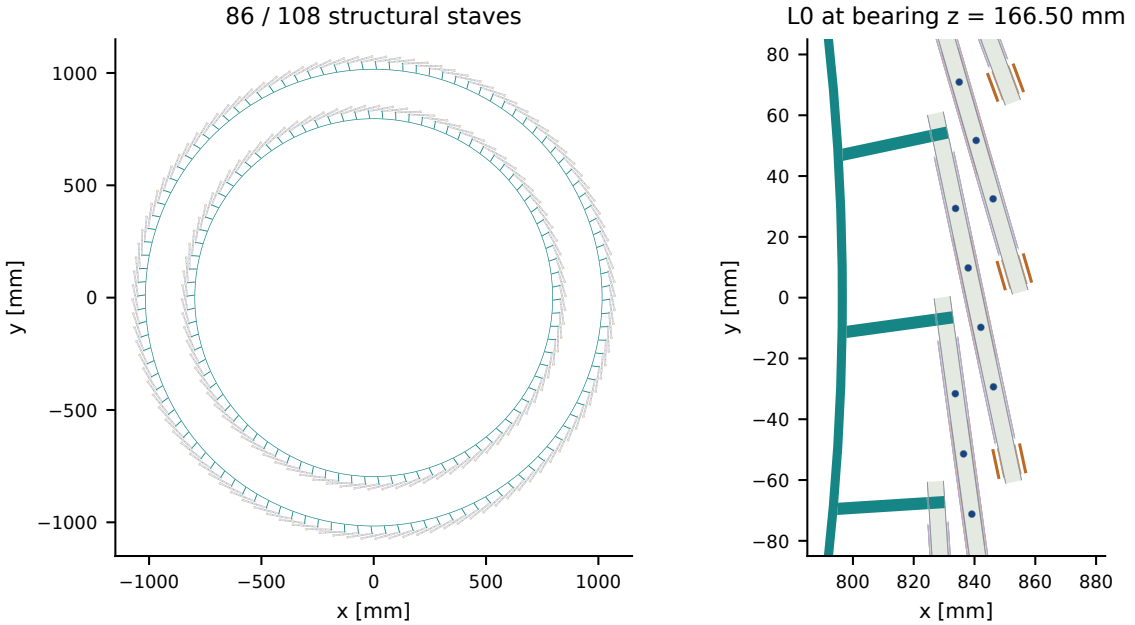


Figure 2.14. True long-strip barrel section at global $z = 166.50$ mm, inside a bearing station. Left: both layers. Right: the innermost stave, bearing ring and local mounting connection at equal transverse scales. All shapes are intersections of the retained solids, including the chord of the potted transverse pin. Purple is sensitive silicon, teal the bearing and mount, grey structural and thermal components, and gold local electrical services.

The allocated normal displacement is 50 µm, divided into 40 µm gravity deformation and a 10 µm joint allowance. Fourteen stations give 42.25 µm gravity deformation and fail the allocated gravity budget; fifteen give 35.49 µm and are the minimum under this screen. **17 stations per stave**, at 163.75 mm spacing from $z = -1310$ to +1310 mm, including the centre, are realised.

At 17 stations the bounded result is 25.95 µm, comprising 5.06 µm bending, 20.67 µm core shear and a small overhang allowance. An independent continuous Timoshenko model gives 23.46 µm, with the two retained mesh spacings agreeing within 0.1 µm. Adding the joint allocation leaves margin within the 50 µm fixture goal. Core shear dominates at this spacing; crediting only skin bending would understate the required support count. Actual bond and fastener compliance, ring/global-frame deformation, local indentation, torsion, vibration and thermal bow require independent qualification.

2.4.4 Barrel cooling and cable routes

Each half-stave has two U-circuits serving nine complete pairs each. Four straight titanium legs at $u = -30, -10, +10, +30$ mm use 2.5 mm outside diameter and 0.14 mm wall. Their separated returns have 10 mm centre radius near $|z| = 20$ mm. Pipe envelopes are explicitly removed from the foam. Supply and return emerge at the outer signed stave end. Both sensor faces connect through insulated local buses and end boards to two harnesses per half-stave; harness assignment counts complete pairs once.

The conductor-sized cable fixture allocates a 6 mm LV/HV bundle and a 3.6 mm bundle containing 48 coated fibres. The power envelope contains 6.14 mm^2 copper and 5.24 mm^2 insulation per unit length. These conductor and volume hypotheses are distinct from the larger gross cable comparator used for short strips. The 11 V, 88.8 W, 4.1 m comparator

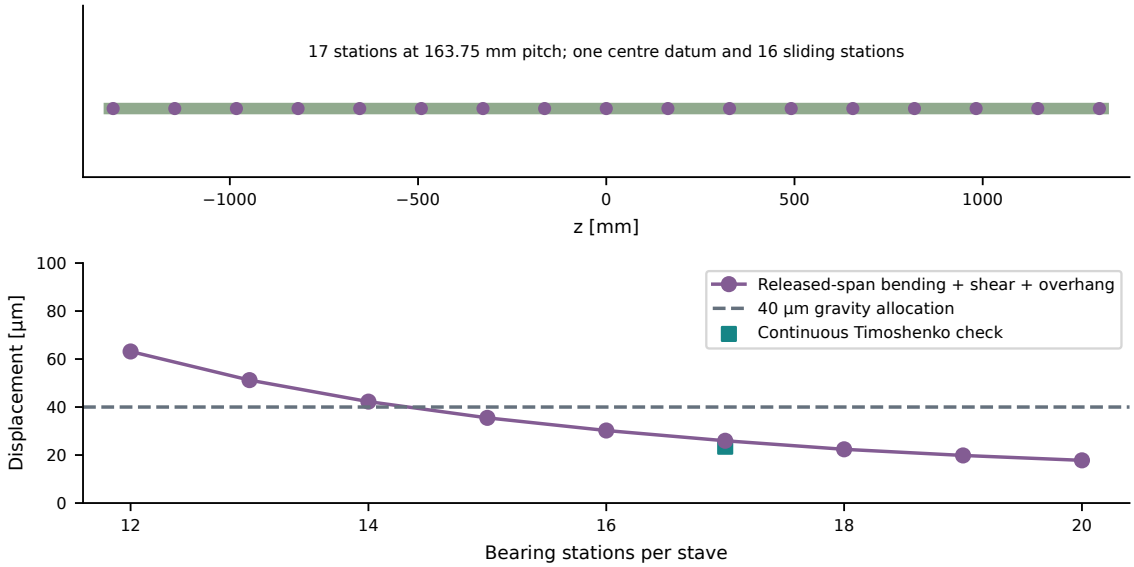


Figure 2.15. Selected long-strip bearing schedule and retained normal-load screen. The released-span estimate includes bending, foam shear and overhang allowance. The separate continuous Timoshenko solution is shown at 17 stations. The curve uses the stated payload and elastic hypotheses; it gives no qualification of the actual joints, rings, thermal bow or torsion.

gives loop/return drops of 0.397 V/0.199 V; increasing resistivity by a factor 1.4 fails the return criterion. Dielectric, terminations, leakage and operational power still need a component-level specification.

The barrel collector occupies $r = 800$–1169 mm, $|z| = 1325$–1435 mm. Its downstream trunk uses the fixed $r = 1144$–1169 mm corridor to $|z| = 3\,500$ mm. Cable and pipe turns reserve 50 mm and 25 mm centre radii. These are clearance reservations, without qualified elbows, weld access, manifold balancing or connector installation. The barrel-only conductor-sized packing cases pass; the larger 13.4 mm gross power comparator does not. Endcap demand must subsequently be included in the same allocation.

2.4.5 Endcap petals, mounting and stereo placement

The six discs per end use the long-strip datums in Table 2.10. The nominal active annulus is $r = 786.46$–1108.33 mm. Four rings contain 72, 72, 84, 84 complete pairs, or 312 pairs per disc and 26 per petal. Twelve petals per disc give 3 744 pairs and 7 488 faces across both ends. Each pair keeps the tangential/radial reference axes and physical ±20 mrad stereo rotations.

The petal uses the same 5 mm foam core, with 0.10 mm glue and 0.30 mm skins. Four full-height integral CFRP webs replace corresponding foam volumes; they are part of the carrier, not an added backing plate. Four outward elevations give sensor-face separations of 6.9, 9.9, 12.9, 15.9 mm. The carrier spans $r = 784$–1130 mm. Inner and outer bearing rings span 783.5–785.5 and 1130–1140 mm. The inner interface leaves only 0.5 mm to the short-strip service boundary and needs a detailed clearance/tolerance review.

Potted lands and keyed shoes carry load to three contacts. One outer contact fixes in-plane position, the other slides tangentially, and the inner key slides radially. The ±0.3 mm motion provision avoids redundant rigid thermal constraints. The proposed

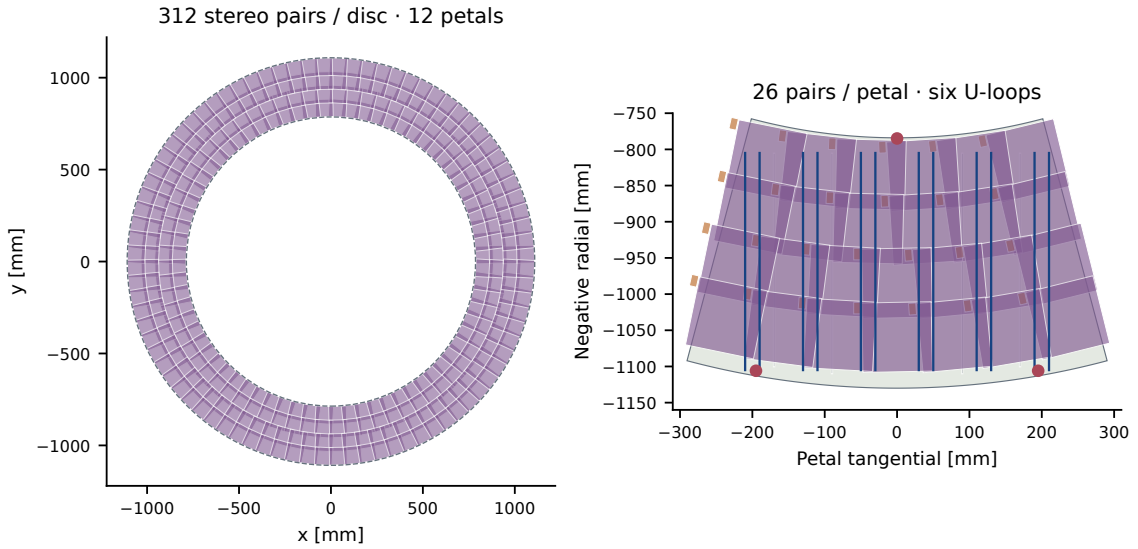


Figure 2.16. Long-strip endcap. Left: one sensor face of each pair on the first positive disc. Right: one continuous structural petal, with its four integral webs, twelve straight legs of six cooling circuits and three mounting contacts. All depths are projected. Blue marks tubes, grey integral webs and red the mounting contacts. Both sensors of each pair have distinct physical stereo frames; only one face is drawn to expose the carrier and routing.

contacts, narrow inner key and grounded rings require plate/joint/global-frame FEA and cold-cycle testing.

The worst complete petal payload is about 1.5 kg. At assumed $E = 70$ GPa and $G = 5$ MPa, the $0.25g$ axial fixture gives 43.85 µm including the 10 µm joint allocation. A horizontal handling load of $1g$ gives 145.41 µm and fails even the tested stiffer combination. A temporary handling frame is therefore recommended. This beam proxy assumes distributed rim reactions; three discrete contacts, plate twisting, local indentation and in-plane gravity cannot be qualified by that result.

2.4.6 Petal cooling, dielectric and cumulative services

Six U-loops per petal distribute twelve straight legs from local radius 805 to 1 105 mm. Tube size and 10 mm returns match the barrel. Nearest-leg assignment gives $4, 4, 6, 4, 4, 4$ complete pairs per circuit, counting both faces as one heat-load unit. The integral webs stop before $r = 1\,114$ mm; outer collection buses start at 1 115 mm. Core holes and bus cutouts are disjoint, conserving the remaining structural material.

The final 0.05 mm copper foil has 0.15 mm polyimide around every exterior face. The spine and outer collector share their intended conductor junction; dielectric is partitioned around that joint so it is neither a bare-carbon contact nor a double-subtracted foam region. This geometric correction does not qualify breakdown voltage, radiation endurance or termination practice.

Three outer harnesses per petal serve groups of $12, 12, 2$ pairs. They reach a $r = 1100$–1169 mm fan, at local outward offsets 20–130 mm, before the longitudinal trunk. The cumulative endcap service inventory includes the pinned barrel loads and every upstream disc; it replaces the standalone barrel corridor. The final reference demand is 3 469.83 mm^2 per sector against 5 449.88 mm^2 capacity.

The farthest pair centre is 30 mm from a pipe leg. At 3.7 W per face, normal foam-slab lower bounds are small, but a pessimistic single lateral path gives temperature rises of 26.43 – 105.71 K over the tested conductivity range and fails all temperature fixtures. The lower bound omits spreading/contact resistance; the lateral case omits beneficial parallel paths. Neither is a predicted sensor temperature. Anisotropic thermal analysis, qualified graphite-to-pipe bridges or a revised pipe distribution must establish the operating margin. Enthalpy budgets also omit pressure drop, two-phase stability and dry-out.

2.5 Review baseline and integration priorities

The selected sensor placements, module populations, stereo frames, support datums and fixed service boundaries form the baseline for review. Its intent is to provide a common, reproducible starting point for geometry, engineering and tracking experts. A successful software build or a merged design contribution does not close a failed engineering screen or establish human design sign-off.

The local models have substantial construction and transport evidence. The remaining work is concentrated at coupled interfaces, where independently reasonable subsystems can still conflict. The review should address Table 2.11 before freezing an integrated detector release.

Table 2.11. Cross-system review priorities and the evidence needed to close them. Existing failed cases remain controls for subsequent revisions.

Interface	Required review outcome
Acceptance and transitions	Combined finite-plane hit maps and hit multiplicity over the luminous region, both charges and the specified momentum range; retain old-datum and original-annulus losses. Establish efficiency with material, response and reconstruction.
Common mechanics	Qualify global cages, ring joints, petal keys and thermal motion. Confirm the 17-station long-stave recommendation with measured elastic properties, joint compliance and global-frame loads. Provide handling and installation tooling, especially for long-strip petals.
Cooling and thermal contacts	Couple realistic electronics/leakage power, anisotropic spreading and contact resistance to pressure drop, flow sharing and two-phase stability. Resolve short-petal thermal failures and the long-petal lateral-path screen, retaining warm and adverse cases.
Power, data and routing	Specify actual sensors/ASICs, harnesses, insulation, terminations and bandwidth. Resolve short-strip nominal trunk overfill and radial turns, long-strip adverse packing and warm return drop, and pixel trunk/flange packing. Provide strain relief and access within fixed limits.
Integrated material model	Replace duplicated standalone service cells once; audit all shared volumes and interfaces together. Produce full directional radiation/interaction-length maps and a constituent inventory before quoting a tracker material budget or mass.
Measurement and alignment	Preserve true sensor axes and IDs, qualify strip pair transport to a common plane, and define response, survey and alignment covariances. Validate conversion/navigation before interpreting fitted-track resolution.

The immediate recommendation is to review the selected layouts and local load paths together, then resolve common services and barrel/endcap transitions in one integrated model. This order prevents a subsystem fix from consuming space already allocated to another subsystem. Fixed envelopes and surviving sensor identities should remain comparison controls; proposed revisions must report their material, acceptance and engineering consequences explicitly.

The report sources retain hashed data snapshots, exact input/producer provenance, actual native execution identities and reproducible figure recipes. Compilation uses

checked-in vector figures and tables only. Regeneration does not rerun simulation or change geometry. The reviewed publication snapshot is therefore distinct from the native validation state and from any future accepted detector release.

Strip detector references

[1] ATLAS Collaboration, *Expected Tracking Performance of the ATLAS Inner Tracker at the High-Luminosity LHC*, JINST 20 (2025) P02018, arXiv:2412.15090v2. Section 2.2, Table 1: barrel azimuthal tilts and their clearance/overlap rationale. Versioned public article.

[2] G. D'Amen et al., *First stave results towards mitigating sensor fracturing with interposers in the ATLAS ITk strips barrel*, JINST 20 (2025) P09025, arXiv:2508.18015v1. Sections 1–2: cold mechanical stress and interposer mitigation. Versioned public article.

[3] S. H. Abidi et al., *Cracking under pressure—investigating mitigation approaches for silicon fractures on ATLAS strip tracker petals at cold temperatures*, JINST 21 (2026) P07026, arXiv:2607.25080v1. Section 1: cooled carbon petals, titanium tubes and copper/polyimide buses; §1.1: cold-fracture and differential-expansion risks. Versioned public article.